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Canadian Math Kangaroo Contest

Part A: Each correct answer is worth 3 points

1. Which cloud contains only numbers less than 7?

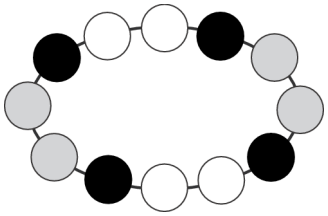
(A)

(B)

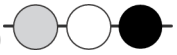
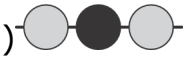
(C)

(D)

(E)



2. Which figure shows a part of this necklace?

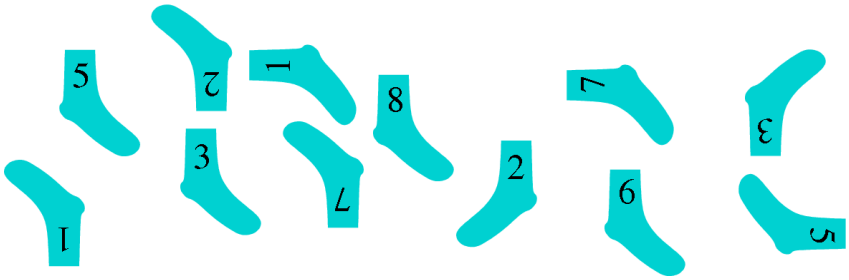
(A) 
(B)
(C)
(D) 
(E)

3. There are 12 children in a line. Lucy is the 7th from the front and Kim is the 2nd from the back. How many children are there between Lucy and Kim?

FRONT

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

4. Jorge pairs his socks so that the numbers match. How many pairs can he make?



(A) 3 (B) 4 (C) 5 (D) 6 (E) 8

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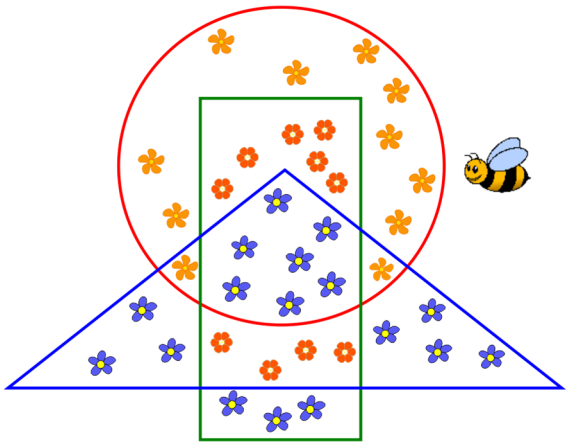
5. If tomorrow is Tuesday, what day was yesterday?

(A) Sunday (B) Thursday (C) Wednesday (D) Monday (E) Saturday

6. Maya Bee was gathering pollen from all of the flowers that lie inside the rectangle, but are outside the triangle. From how many flowers did she collect pollen?

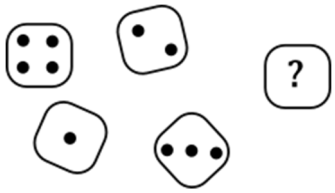


(A) 6 (B) 9 (C) 13 (D) 17 (E) 20

<u>Part B: Each correct answer is worth 4 points</u>

7. Jutta throws five regular dice. The total number of dots is 15. Four dice are as shown on the right.

How many dots must there be on the fifth die?



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

8. Clara wrote a hidden message to her friend by replacing each letter, as shown:

R	I	N	G		K	A	R	L	O
*	@	#	$		!	%	*	?	+

How would RANGO be written?

(A) *#%$+ (B) *%#$+ (C) *%$#+ (D) *@#$+ (E) *+$#+

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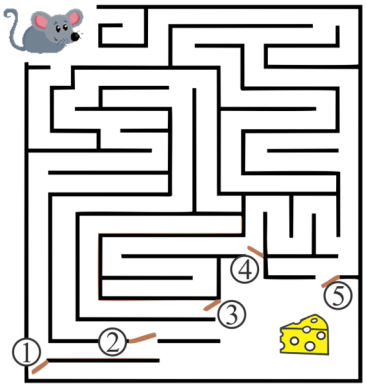
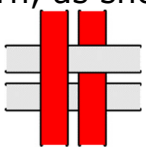
9. You have to close two of the five gates so that the mouse cannot reach the cheese.

Which gates should you close?

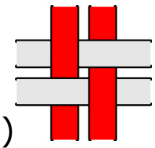
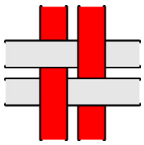
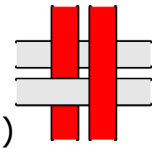
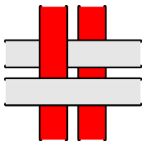
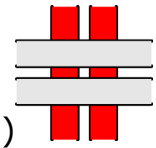
(A) 1 and 2
(B) 2 and 3
(C) 3 and 4
(D) 3 and 5
(E) 4 and 5



10. Four strips are woven into a pattern, as shown.

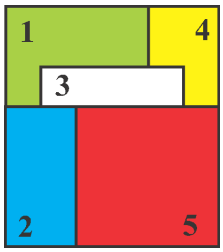


What do you see when you look at it from the other side?

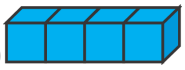
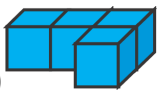
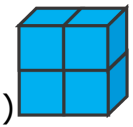
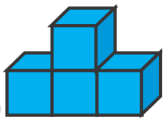
(A)  (B)  (C)  (D)  (E) 

11. Five square cards are stacked on a table, as shown on the right. The cards are removed one by one from the top of the stack. In what order are the cards removed?

(A) 5-2-3-1-4 (B) 5-2-3-4-1 (C) 4-5-2-3-1
(D) 5-3-2-1-4 (E) 1-2-3-4-5



12. Each of the shapes shown is made by gluing together four cubes of the same size. The shapes are to be painted. Which shape has the smallest area to be painted?

(A)  (B)  (C)  (D) (E) 

<u>Part C: Each correct answer is worth 5 points</u>

13. On a farm, there are only sheep and cows. The number of sheep is 8 more than the number of cows. The number of cows is half the number of sheep. How many animals are on the farm?

(A) 16 (B) 18 (C) 20 (D) 24 (E) 28

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14. Here are nine squares: . First, Andrea replaced all the black squares with white ones. Next, Bob replaced all the grey squares with black ones. Finally, Chris replaced all the white squares with grey ones. What did they get at the end?

(A)

(B)

(C)

(D)

(E)

15. Peter chose a square of four cells in the table so that the sum of the four numbers inside the square is greater than 63. Which of the following numbers must be in the chosen square?

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

(A) 14 (B) 15 (C) 17 (D) 18 (E) 20

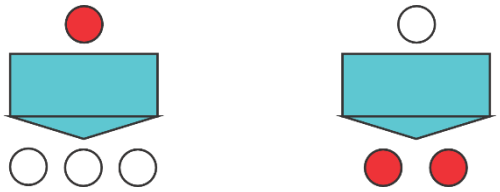
16. There were sparrows on two trees. At first, seven sparrows flew from Tree 1 to Tree 2. Then, four sparrows flew from Tree 2 to Tree 1. After that, there were five sparrows on each tree. How many sparrows were sitting on Tree 1 to begin with?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

17. The age difference between Adam and his older sister Lucy is 7 years. Adam is 8 years younger than Monica. Which of the following answers shows the three children, from the oldest to the youngest?

(A) Adam, Lucy, Monica (B) Lucy, Monica, Adam (C) Monica, Adam, Lucy
(D) Adam, Monica, Lucy (E) Monica, Lucy, Adam

18. Amalia's machine converts one red token into three white tokens and one white token into two red tokens.



Amalia has three red tokens and one white token: .

She uses the machine three times to convert three of her tokens.

What is the smallest number of tokens she can end up with?

(A) 7 (B) 6 (C) 8 (D) 5 (E) 9

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